

Zhanwei Zhang (张展玮)

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No.1 Du Xue Rd, Nansha District, Guangzhou, Guangdong

Education

M.Phil. in Data Science and Analytics, Hong Kong University of Science and Technology (Guangzhou) Sep 2025 ~ Present

Information Hub

Advisor: Prof. Zishuo Ding

B.Sc. in Computer Science and Technology, Southern University of Science and Technology (SUSTech) Sep 2021 ~ Jun 2025

Turing Master Class

Advisor: Prof. Yepang Liu

GPA: 3.79 / 4.0 | Weight Avg Score: 90.92 | Ranking: 36 / 195

Main courses: Mathematical Logic (A+), Java Programming (A+), Data Structures & Algorithms (A), Machine Learning (A),

Database Systems (A-), Compilers (B+)

Internships

Large Language Model Intern in Lingsome, Shenzhen

Aug 2024 ~ Aug 2025

Integrated multi-type Retrieval Augmented Generation (RAG) and GraphRAG systems.

Optimized GraphRAG code to extract better entities & relationships and construct domain-specific knowledge graphs.

Developed and refined pipelines to extract improved entities and relationships.

Visiting Researcher in Wuhan University, Wuhan

May ~ Aug 2024

Advisor: Prof. Jinfu Chen (WHU); Prof. Weiyi Shang (UWaterloo)

Focused on software logging and failure workarounds.

Developed an automated analysis pipeline to extract, filter, and sample code commits containing try-catch blocks.

Selected Projects

WiTH - AI Health Companion

Jun 2026

Built an AI health companion for the 2026 Shenzhen Hackathon, combining health records, AI chat, and personalized reminders.

Designed the product workflow from daily health tracking to companion-style interaction, helping the team win Third Prize.

AI Micro-Drama Studio

Sep 2025 ~ Jan 2026

Built an end-to-end pipeline that transforms novels or scripts into production-ready storyboards.

Unified creation, generation, editing, and export in one interface to streamline publish-ready video production.

Customizable AI Companion Doll (OpenHarmony TSC Project)

Sep 2025 ~ Jan 2026

Developed an AIoT system integrating cloud-based LLMs to provide personalized companionship through customizable personalities and multimodal interaction.

Built the cross-platform frontend, including a mobile App and Mini-program.

Othello Game through Java and Python Programming with Strong AI

Oct ~ Dec 2021 & Mar 2023

Developed visually appealing interface and implemented Monte Carlo & Alpha Beta Pruning algorithm.

Canteen Traffic Monitoring

Dec 2023 ~ Jan 2024

Calculated the length of the queue by monitoring data and displayed a chart showing the changes in queue length.

Won award for finalist in National College Students' Innovation and Entrepreneurship Training program.

About 30,000 visits within three months.

Simple Compiler

Sep 2023 ~ Jan 2024

Developed a compiler that translates C language files into Intermediate Representation (IR) and MIPS32.

Supported essential features such as I/O operations, control flow and function calls.

Research

Numerical Error Detection in Floating-Point Computing

Sep 2024 ~ Feb 2026

PI-detector: A condition-number-guided perturbation approach can replace costly high-precision oracles, finding 173/174 significant-error cases at about 0.13% of oracle cost (up to $73.46\times$ faster).

MGDE: Turn detection into a Newton-Raphson-guided convergent input search, reaching 80 bugs / 47 functions (vs. 70/46 ATOMU, 53/42 FPCC) while being $41.71\times$ and $10.17\times$ faster, respectively.

Reproduction and Evaluation of R1-style Reasoning Pipeline

May 2025 ~ Jun 2025

Replicated the DeepSeek-R1 training pipeline by implementing GRPO and Cold-start SFT, significantly enhancing the multi-step reasoning capabilities of Qwen2.5 models (0.5B to 7B).

LLM-Based JSON Parser Fuzzing for Bug Discovery and Behavioral Analysis

Sep 2023 ~ Jan 2024

Used opensource LLMs such as Llama2-7B/13B to generate test cases.

13 JSON Parsers and over 100 types of cases have been tested. Over 26 behavioral diversities have been found.

Publications

Tan, Y., **Zhang, Z.**, Ding, Z., Zheng, L., Chen, J., & Shang, W. (2025). *A mathematics-guided approach to floating-point error detection*. arXiv. <https://arxiv.org/abs/2510.10081> (Under Review)

Tan, Y., **Zhang, Z.**, Chen, J., Ding, Z., Xuan, J., & Shang, W. (2025). *Computing floating-point errors by injecting perturbations*. arXiv. <https://arxiv.org/abs/2507.08467>

Han, Y., Shen, H., He, X., Mai, Z., Zhang, R., Zheng, Z., Liu, Y., Zhang, X., Li, G., **Zhang, Z.**, Liang, Z., Chen, Y., Xie, Y., Li, M., Shen, G., Wang, C., Ye, J., Zhu, L., Fu, T.-M., & Yang, X. (2025). A comprehensive analysis of interflight variability in carbon dioxide emissions from global aviation. *Environmental Science & Technology*, 59(12), 6179-6191. doi:10.1021/acs.est.5c02371

Patents

一种点餐方法、系统、终端及介质 (Innovative Ordering Method, System, Terminal, and Medium Patent)

May 2023

Innovated a method and system to alleviate peak-hour traffic in cafeterias.

Applied on May 5, 2023; Application no: 202310498065

Skills

Languages: English (Fluent; IELTS: 6.5), Mandarin (Native), Cantonese (Native)

Programming Languages & Frameworks: Java, Python, C/C++, SQL, Spring Boot, Vue, React

Tools: IntelliJ IDEA, PyCharm, Visual Studio Code, Anaconda, Git, CMake

Honors & Scholarships

Third Prize, Shenzhen Hackathon

Jun 2026

Special Innovation Award (Unique Winner) & Second Prize, OpenHarmony Competition Training Camp

Sep 2025

Postgraduate Studentship (PGS), HKUST(GZ)

Sep 2025

Outstanding Student, SUSTech

Jan 2024

Honorable Mention, Mathematical Contest in Modeling

May 2023

Finalist, National College Students' Innovation and Entrepreneurship Training program

Jun 2023

Third Prize, China Undergraduate Mathematical Contest in Model

Sep 2023